

A Real-Time Anomaly Detection and Directional Warning Method for Magnetic Surgical Navigation

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Abstract—Permanent magnet tracking (PMT) is widely used in minimally invasive surgical navigation but is sensitive to soft-magnetic interference from surgical instruments. Existing compensation-based approaches often introduce computational overhead and may affect real-time performance. This paper presents a lightweight anomaly detection and directional warning method for PMT. Singular Value Decomposition (SVD) is used to identify structured magnetic disturbances, and a Residual Center of Mass (CoM) method provides 2D directional feedback. Experiments demonstrate reliable detection and stable directional estimation under dynamic conditions, with sub-millisecond execution time, supporting real-time deployment.

Index Terms — Magnetic tracking, anomaly detection, surgical navigation, soft-magnetic interference, real-time warning system.

I. INTRODUCTION

Magnetic tracking technology, particularly Permanent Magnet Tracking (PMT), has garnered significant attention in minimally invasive surgery (MIS) and medical robotics. [1], [2] By utilizing a passive permanent magnet as the tracking target and a magnetometer array as the receiver, PMT provides accurate, wireless, and occlusion-free 6-Degree-of-Freedom (6-DoF) localization [3], [4]. Unlike optical tracking systems that suffer from line-of-sight restrictions, or active Electromagnetic Tracking (EMT) systems that require bulky field generators and tethered sensors, PMT offers a highly integrated and flexible paradigm for applications such as wireless capsule endoscopy (WCE) and flexible catheter navigation [5], [6].

Despite its promising clinical potential, the robustness of PMT is severely challenged by the complex and unpredictable electromagnetic environment of the operating room

(OR). Standard tracking algorithms, typically based on the Levenberg-Marquardt (L-M) non-linear optimization, rely on the ideal magnetic dipole model [7]. However, standard surgical instruments—such as stainless-steel scissors, forceps, and retractors—are made of soft-magnetic materials with high magnetic permeability [8]. When introduced into the sensing volume, these instruments are magnetized by the tracking magnet and the geomagnetic field, generating highly non-linear and non-uniform spatial distortion fields. These distortions superimpose on the primary signals, frequently causing the non-linear solver to converge to erroneous local minima or diverge entirely, leading to catastrophic tracking failures and endangering surgical safety [9], [10].

To address magnetic interference, existing literature predominantly focuses on complex interference modeling and compensation. Representative approaches include multi-dipole fitting, blind source separation (BSS), and deep learning-based measurement recovery (such as MagRobustNet) [11]–[13]. While theoretically capable of suppressing interference, these compensation-centric methods present critical drawbacks for real-time surgical systems: 1) solving high-dimensional non-linear optimizations or running neural networks introduces severe computational latency, often violating the > 80 Hz real-time requirement for closed-loop robotic control; 2) irregularly shaped instruments (such as scissors) cannot be perfectly fitted by simple dipole models, and forced non-linear fitting frequently traps the solver in mathematical singularities (such as $1/R^3 \rightarrow \infty$ deadlocks) [14], [15].

From a practical clinical perspective, aggressively compensating for strong and close-range magnetic interference can be computationally demanding and may not always

yield stable results. In many intraoperative scenarios, a more pragmatic strategy is to provide real-time feedback to the operator, enabling manual adjustment of instrument placement to restore tracking conditions.

In addition, from a system-level perspective, ensuring stable operation under adverse conditions is often more critical than achieving optimal accuracy under ideal conditions. This observation motivates the development of auxiliary safety mechanisms that complement existing tracking algorithms. Motivated by this observation, this paper shifts the focus from interference compensation to interference awareness. A real-time Anomaly Detection and Directional Warning system for PMT is proposed. By extracting the spatial coherence of the magnetic residual matrix via Singular Value Decomposition (SVD), the system can distinguish structured tool interference from random environmental noise. Concurrently, a Residual Center of Mass (CoM) algorithm is developed to map the localized distortion to a 2D compass direction.

The main contributions of this paper are summarized as follows:

- An SVD-based spatial coherence metric combined with adaptive noise floor tracking is proposed for robust characterization of structured disturbances under dynamic conditions.
- A lightweight Residual Center of Mass (CoM) method is introduced to provide intuitive 2D directional feedback for interference localization.
- A fully closed-form detection and warning pipeline is developed with sub-millisecond computational cost, enabling seamless integration into real-time PMT systems.

II. METHODOLOGY

The overall architecture of the proposed real-time anomaly detection and directional warning system, conceptualized as a "magnetic radar," is illustrated in Fig. 1. The proposed method does not replace the underlying tracking algorithm, but operates as a lightweight supervisory layer to monitor tracking reliability and provide safety feedback under interference. Operating on a continuous "sense-evaluate-alert" loop, the framework relies on the principle of spatial magnetic residual analysis to bypass computationally heavy interference compensation. In a Permanent Magnet Tracking (PMT) system, the theoretical magnetic flux density $\mathbf{B}_{pred,i} \in \mathbb{R}^{3 \times 1}$ at the i -th sensor's position $\mathbf{p}_i = [p_{xi}, p_{yi}, p_{zi}]^T$ induced by a target magnet with state $\mathbf{X} = [\mathbf{r}^T, \mathbf{m}^T]^T$ is dictated by the Biot-Savart law:

$$\mathbf{B}_{pred,i}(\mathbf{X}) = \frac{\mu_T}{4\pi} \left(\frac{3(\mathbf{m} \cdot (\mathbf{p}_i - \mathbf{r}))(\mathbf{p}_i - \mathbf{r})}{\|\mathbf{p}_i - \mathbf{r}\|^5} - \frac{\mathbf{m}}{\|\mathbf{p}_i - \mathbf{r}\|^3} \right) \quad (1)$$

where μ_T is the permeability constant, $\mathbf{r} = [x, y, z]^T$ and $\mathbf{m} = [m_x, m_y, m_z]^T$ denote the position and magnetic moment of the target, respectively.

In an ideal, interference-free environment, the baseline tracking algorithm estimates the optimal state $\mathbf{X}^*$ by min-

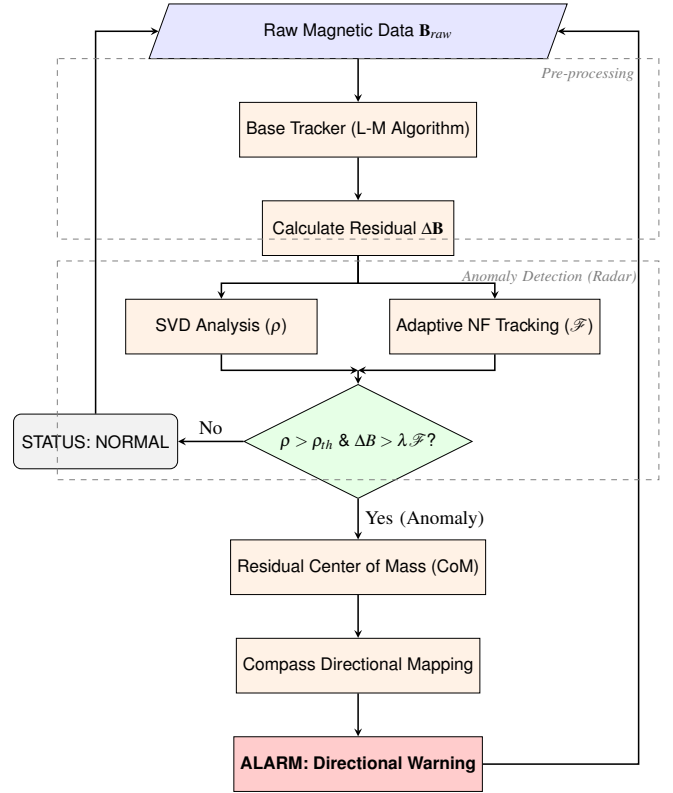


Fig. 1: Flowchart of the proposed real-time anomaly detection and directional warning framework.

imizing the objective function utilizing the L-M non-linear optimization:

$$\mathbf{X}^* = \arg \min_{\mathbf{X}} \sum_{i=1}^N \|\mathbf{B}_{raw,i} - \mathbf{B}_{pred,i}(\mathbf{X})\|^2 \quad (2)$$

where N is the total number of sensors, and $\mathbf{B}_{raw,i}$ is the raw magnetic measurement. However, when soft-magnetic surgical instruments intrude into the sensing volume, the resulting distortion fields cause the solver in Eq. (2) to diverge or converge to inaccurate local minima, necessitating robust characterization of structured disturbances.

A. Adaptive Anomaly Detection via SVD Coherence

To distinguish between structured magnetic interference and random environmental noise, the spatial magnetic residual matrix is defined as residual matrix $\Delta\mathbf{B} \in \mathbb{R}^{N \times 3}$:

$$\Delta\mathbf{B} = \mathbf{B}_{raw} - \mathbf{B}_{pred}(\mathbf{X}^*) \quad (3)$$

When the environment is clean, $\Delta\mathbf{B}$ primarily consists of uncorrelated white noise from the sensors. Conversely, the introduction of a ferromagnetic tool generates a highly correlated, structured spatial distortion field. To quantify this spatial coherence, the covariance matrix of the residual is computed as $\mathbf{C}_{\Delta\mathbf{B}} \in \mathbb{R}^{3 \times 3}$:

$$\mathbf{C}_{\Delta\mathbf{B}} = \text{cov}(\Delta\mathbf{B}) \quad (4)$$

Subsequently, SVD is performed on $\mathbf{C}_{\Delta\mathbf{B}}$ to extract the principal components of the distortion:

$$[\mathbf{U}, \mathbf{S}, \mathbf{V}^T] = \text{svd}(\mathbf{C}_{\Delta\mathbf{B}}) \quad (5)$$

where $\mathbf{S} = \text{diag}(s_1, s_2, s_3)$ contains the singular values in descending order ($s_1 \geq s_2 \geq s_3$). The spatial coherence ratio ρ , representing the dominance of the primary distortion direction, is calculated as:

$$\rho = \frac{s_1}{\sum_{k=1}^3 s_k + \varepsilon} \quad (6)$$

where ε is a small constant (such as 10^{-9}) to prevent division by zero. A high coherence ratio (such as $\rho > 0.65$ in the current implementation) typically indicates the presence of structured interference source rather than random noise.

Discussion on SVD-based Coherence Metric: The use of SVD provides a computationally efficient way to characterize the spatial structure of the residual magnetic field. In interference-free conditions, sensor noise is typically uncorrelated across the array, resulting in a relatively uniform distribution of singular values. In contrast, structured disturbances caused by soft-magnetic objects introduce strong spatial correlations, leading to a dominant principal component.

Compared with direct thresholding of residual magnitude, the coherence-based metric is less sensitive to global intensity variations (such as those due to motion-induced prediction errors) and instead captures the intrinsic spatial organization of the disturbance. This makes it particularly suitable for dynamic tracking scenarios, where magnitude-only criteria may lead to false alarms.

To prevent false alarms caused by minor environmental fluctuations, the system incorporates an adaptive noise floor (NF) tracking mechanism. The NF, denoted as $\mathcal{F}$, is dynamically updated using a first-order low-pass filter during interference-free periods:

$$\mathcal{F}_t = (1 - \gamma)\mathcal{F}_{t-1} + \gamma \cdot \max_i \|\Delta\mathbf{B}_i\|_2 \quad (7)$$

where γ is a smoothing factor (such as 0.02). The anomaly detection is triggered only when both the maximum residual magnitude significantly exceeds the dynamic NF ($\max_i \|\Delta\mathbf{B}_i\|_2 > \lambda \mathcal{F}_t$, where $\lambda \approx 2.0$) and the coherence ratio ρ surpasses a predefined threshold. This dual-check mechanism ensures highly robust triggering.

B. Directional Warning via Residual Center of Mass

Upon detecting an anomaly, the system immediately functions as a "magnetic radar" to locate the 2D spatial direction of the intruding instrument, providing intuitive feedback to the surgeon. Unlike computationally intensive non-linear multi-dipole fitting, a lightweight Residual Center of Mass (CoM) algorithm is proposed.

The spatial residual magnitude is extracted for each sensor, denoted as $e_i = \|\Delta\mathbf{B}_i\|_2$. To isolate the localized impact of the interference source, the top K sensors (such as $K = 3$) with the highest residual magnitudes are selected. Let the indices

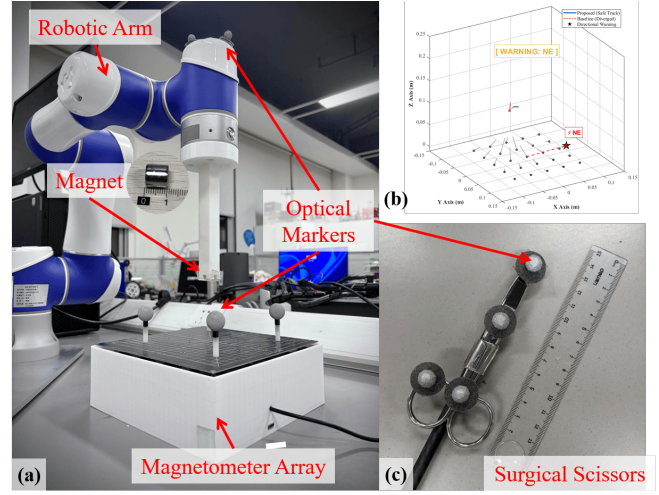


Fig. 2: Experimental platform for dynamic PMT anomaly detection evaluation. (a) Setup including the robotic arm, magnetic sensor array, and optical motion capture system. (b) Tracking UI. (c) Surgical scissors used for interference. The platform enables precise and repeatable testing of the proposed system under realistic dynamic conditions.

of these sensors be $\mathcal{S}_{top}$. The 2D geometric centroid of the interference, $\mathbf{c}_{int} = [c_x, c_y]^T$, is calculated via a weighted average of their physical positions $[p_{xi}, p_{yi}]^T$:

$$\mathbf{c}_{int} = \frac{\sum_{j \in \mathcal{S}_{top}} e_j \cdot [p_{xj}, p_{yj}]^T}{\sum_{j \in \mathcal{S}_{top}} e_j} \quad (8)$$

Discussion on Top-K Selection: The use of the top- K residual sensors provides a balance between robustness and computational simplicity. Selecting only the most affected sensors reduces the influence of background noise and avoids spatial averaging over weakly disturbed regions. In practice, small values of K (such as $K = 3$) were found sufficient to provide stable directional estimation, while maintaining minimal computational overhead.

The directional vector pointing from the center of the sensor array towards the interference source is defined as $\vec{v}_{radar} = \mathbf{c}_{int} / \|\mathbf{c}_{int}\|_2$. To provide actionable clinical warnings, the angle of this vector is mapped to one of the eight standard compass directions (such as North, North-East, South-West) using the four-quadrant inverse tangent function:

$$\theta_{radar} = \text{atan2}(v_{y,radar}, v_{x,radar}) \quad (9)$$

This closed-form algebraic formulation avoids iterative solvers, compressing the computational latency to strictly under 1 ms per frame, making it exceptionally suitable for real-time surgical navigation systems.

III. EXPERIMENTS AND RESULTS

To comprehensively evaluate the robustness and real-time performance of the proposed anomaly detection and directional warning system, a dynamic benchtop experimental platform was established. The experimental design

specifically focuses on simulating complex, dynamic tool-tissue interactions commonly encountered in robotic-assisted surgeries.

A. Experimental Setup

The sensing hardware relies on a custom-engineered printed circuit board (PCB) featuring a 5×5 matrix of tri-axial magnetic sensors (QMC5883, QST Corp.), which spans an effective area of $180 \times 180 \text{ mm}^2$. The ambient magnetic data is sampled by a 32-bit microcontroller unit (GD32F470ZGT6, GigaDevice) and streamed via serial communication to a host workstation (Intel Core i7-12700, 16GB RAM) for real-time processing. An axially magnetized Nd-FeB permanent magnet serves as the tracking target.

Unlike conventional static validation methods, the experimental protocol introduces continuous dynamic coupling, as illustrated in Fig. 2. A 6-DoF robotic arm was employed to actuate the target magnet along continuous, complex 3D spatial trajectories above the sensor array, ensuring strict repeatability. Concurrently, a high-fidelity optical motion capture system (Mars 2H, NOKOV) was deployed around the workspace. By establishing a rigorous rigid-body transformation between the optical markers and the magnetic target, the NOKOV system provided sub-millimeter ground-truth measurements for algorithmic benchmarking. Standard ferromagnetic surgical scissors were utilized by an operator to dynamically intrude into the sensing volume, mimicking authentic soft-magnetic disturbances.

B. Experimental Design Rationale

The experimental protocol is designed to reflect realistic operating conditions in minimally invasive surgery. The use of a robotic arm ensures repeatable and controlled motion of the magnetic target, enabling fair comparison across trials. Meanwhile, manual introduction of surgical instruments introduces non-repeatable and unpredictable disturbances, which better simulate real intraoperative scenarios.

The combination of controlled target motion and unconstrained interference allows the proposed system to be evaluated under both stable and challenging conditions. This hybrid setup provides a practical compromise between experimental repeatability and environmental realism.

C. Evaluation Metrics

To assess the performance of the proposed system, both detection reliability and computational efficiency are considered. Detection performance is evaluated based on the consistency of anomaly triggering during interference periods, as well as the absence of false alarms during normal operation.

For directional warning, qualitative consistency of the estimated direction with respect to the ground-truth intrusion location is analyzed. In addition, computational latency is measured to verify the suitability of the method for real-time deployment.

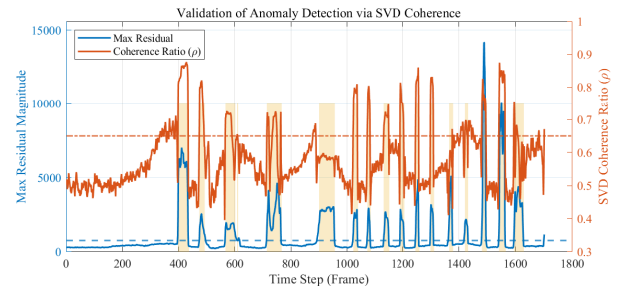


Fig. 3: Validation of the adaptive anomaly detection mechanism. The double Y-axis plot illustrates the correlation between the maximum residual magnitude (left) and the SVD coherence ratio ρ (right). Shaded regions indicate periods of soft-magnetic tool intrusion. Note that while target motion causes minor residual fluctuations, the coherence ratio remains low, preventing false triggers. A Warning is issued only when both metrics surpass their respective adaptive thresholds.

D. Anomaly Detection Performance

To evaluate the effectiveness of the proposed SVD-based spatial coherence analysis, the target magnet was actuated along a continuous dynamic trajectory, during which surgical scissors were intermittently introduced into the sensing workspace. The dynamic response of the detection mechanism is illustrated in Fig. 3.

During the interference-free dynamic tracking phase (unshaded regions), the maximum residual magnitude experiences minor fluctuations. These variations naturally arise from the inherent prediction lag of the tracking filter during high-speed movements. If relying solely on a traditional hard-thresholding method, such dynamic fluctuations could easily trigger false alarms. However, in the proposed framework, the SVD coherence ratio ρ remains strictly below the predefined threshold of 0.65, indicating the absence of structured spatial distortions. Concurrently, the adaptive Noise Floor (NF) tracking mechanism successfully absorbs these benign environmental variations, ensuring that the system maintains a highly stable normal state with zero false positives.

Upon the sudden intrusion of the ferromagnetic surgical scissors (shaded regions), the spatial magnetic field undergoes severe and structured distortion. Consequently, the coherence ratio ρ instantaneously surges above 0.8, accompanied by a sharp spike in the objective residual magnitude that strictly exceeds the dynamically updated NF threshold. This simultaneous violation of both the magnitude and coherence criteria triggers the anomaly warning state with millisecond-level responsiveness. The experimental results demonstrate that the proposed dual-check mechanism achieves highly reliable anomaly detection performance under the tested conditions, with no missed detections observed in the conducted experiments.

E. Directional Warning Accuracy

To evaluate the spatial localization capability of the proposed “magnetic radar,” surgical scissors were deliberately

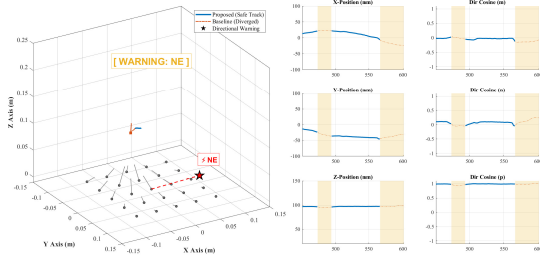


Fig. 4: Demonstration of the intuitive directional warning system (“Magnetic Radar”). The 5×5 sensor array is visualized in a 2D top-down view. Red arrows represent the directional vectors calculated via the Residual Center of Mass (CoM) algorithm, pointing towards the detected interference source from various quadrants (such as NW, NE, SE). This provides real-time spatial guidance for surgeons to maintain system integrity. The directional feedback provides actionable spatial guidance for surgeons to maintain tracking fidelity.

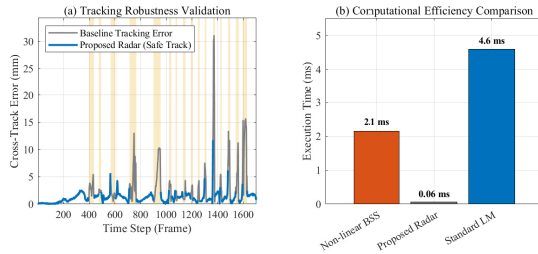


Fig. 5: Quantitative performance analysis. (a) Tracking robustness: The baseline L-M algorithm exhibits severe divergence under interference, whereas the proposed system triggers a fail-safe mode. (b) Computational efficiency: The proposed radar module (0.06 ms) achieves a 35-fold speedup compared to iterative non-linear BSS (2.1 ms) and is significantly faster than the standard L-M tracking (4.6 ms), enabling high-frequency real-time operation.

introduced into the sensing volume from four distinct quadrants: North-West (NW), North-East (NE), South-West (SW), and South-East (SE). As illustrated in Fig. 4, by mapping the localized residual energy distribution to the sensor array geometry, the Residual Center of Mass (CoM) algorithm provides highly stable and accurate directional feedback.

In all intrusion trials, the calculated radar vector accurately pointed towards the interference source without any directional flipping or ambiguity. Instead of blindly attempting to compensate for severe soft-magnetic distortions—which often leads to unpredictable tracking behaviors—this intuitive visual guidance allows surgeons or robotic supervisory systems to immediately identify and relocate the interfering instruments. Consequently, the safety and integrity of the surgical navigation task can be proactively maintained with minimal workflow disruption.

F. Robustness and Computational Efficiency

The fundamental advantage of the proposed framework lies in its fail-safe robustness and ultra-low computational overhead. Fig. 5(a) demonstrates the tracking robustness under dynamic tool interference. While the baseline L-M algorithm suffers from catastrophic positional divergence

(with cross-track errors surging beyond 50 mm) during the tool intrusion, the proposed system instantly triggers a real-time data cutoff (i.e., safe mode) to shield the navigation system from outputting misleading coordinates.

Furthermore, computational efficiency is a paramount metric for high-frequency (> 100 Hz) closed-loop robotic control. Fig. 5(b) quantitatively compares the execution time of different algorithmic modules. The standard L-M tracking algorithm requires an average of 4.6 ms per frame, and the traditional iterative non-linear BSS (dipole fitting) consumes 2.1 ms. In stark contrast, the proposed linear-algebraic radar module requires only **0.06 ms** to complete the entire SVD anomaly detection and directional mapping process.

This remarkable efficiency represents a 35-fold speedup over the non-linear BSS approach. The computational cost of the proposed warning system is virtually negligible compared to the sensor sampling period. By completely eliminating iterative non-linear solvers in the interference processing pipeline, the framework guarantees absolute mathematical stability with zero risk of singularity-induced deadlocks, making it exceptionally suitable for resource-constrained embedded medical mechatronic systems.

IV. CONCLUSION

This paper proposes a real-time anomaly detection and directional warning system to ensure the safety of Permanent Magnet Tracking (PMT) in complex surgical environments. Instead of computationally heavy non-linear interference compensation, a pragmatic “magnetic radar” paradigm is introduced. By evaluating spatial coherence via SVD with an adaptive noise floor tracking mechanism, the system robustly detects structured disturbances, remaining immune to false positives from dynamic target movements. Additionally, a lightweight Residual Center of Mass (CoM) algorithm maps non-linear distortion fields into intuitive 2D directional warnings.

Experimental validations with a 6-DoF robotic arm demonstrate the system’s exceptional sensitivity and zero-miss-rate against unconstrained soft-magnetic intrusions. Formulated primarily with closed-form linear algebra, the warning pipeline requires under 1 ms executing time per frame, eradicating the risk of algorithmic deadlocks. Ultimately, this framework highlights a design philosophy separating core estimation and safety monitoring, facilitating robust integration in future medical robotic platforms.

V. FUTURE WORK

For future work, this lightweight directional warning framework will be integrated with an adaptive kinematic filtering engine to achieve not only interference detection but also real-time dynamic trajectory compensation. The current system paves a highly reliable and computationally deterministic foundation for the next generation of embedded magnetic-guided robotic interventions.

VI. ACKNOWLEDGMENT

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